Alex Doe

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Education

State University College Town, IL

Bachelor of Science in Computer Science; GPA: 3.50 *Aug. 2024 – May 2028*

- Coursework: Operating Systems, Computer Networks, Computer Architecture, Databases, Data Structures & Algorithms **Graduating May 2028; available May – Aug. 2027**

Experience

Acme Labs Austin, TX

Software Engineer Intern *Feb. 2026 – Jun. 2026*

- Owned a search platform end to end across a 5-month rotation, from a multithreaded C++ crawler ingesting 2M+ documents over concurrent TCP connections through an inverted index and a query service holding sub-50ms p99 for 100+ concurrent users, taking every change from design through deployment and monitoring.
- Built the tooling non-engineers actually used: a query-replay harness that reproduced any bad result from captured traffic, and a dashboard letting non-technical stakeholders retune ranking weights without a code change; presented the system to company leadership while working on a multilingual team.

State University, Generative AI Lab College Town, IL

Undergraduate Research Assistant *Jan. 2025 – May 2025*

- Built the real-time inference path for a speech-scoring model in an English-learning application, streaming audio over WebSockets within a 300ms budget, plus the feature pipeline turning 100K+ raw interaction events into the 40+ signals the model consumed.

Projects

Market Data Decoder and Tick Capture Pipeline

C++, Python *Feb. 2026*

- Built a decoder for a simulated exchange's UDP multicast feed parsing 1M+ messages/sec into full-depth order books for 500+ instruments, plus a column-oriented capture format with nanosecond timestamps that cut replay storage 9x and made any session reproducible bit for bit.

Model Validation and Alerting Framework

Python, Kafka, PostgreSQL *Jun. 2026*

- Built a service letting non-engineers define guardrails on a live model's output in a small expression language, evaluating 200+ rules per second against a streaming feed and routing each breach to an alert with the triggering input snapshot attached.

Lock-Free Order Matching Engine

C++, Lock-Free Data Structures *Dec. 2025*

- Implemented a price-time priority matching engine over atomic operations and a ring-buffer message queue, sustaining 500K+ orders/sec at p99 latency under 1μs, validated against 5M+ simulated order events.

Open Source & Research

OpenProjectA and OpenProjectB

Open Source Contributor *Feb. 2025 – Present*

- Contributed production C++ changes to PyTorch core's ATen kernel dispatch across 11 operators with merged BFloat16/Half regression tests, and fixed a soundness bug in OpenAI Triton's MLIR AxisInfo analysis producing incorrect vectorized GPU code, with a FileCheck test merged into CI.

Cache Eviction Behavior Across Key-Value Stores

Solo-Authored, arXiv Preprint *Jul. 2026*

- Recovered the internal cache block sizes of 4 major providers from black-box timing telemetry alone, correcting a widely cited figure by 8x and tracing one provider's run-to-run non-determinism to unpinned load balancing across replicas.

Skills

Languages: C++, Python, Java, Rust, SQL, JavaScript/TypeScript

Systems: Multithreading & Lock-Free Concurrency, TCP/IP & UDP Multicast, Low-Latency Data Pipelines, Performance Profiling

Tools: Linux, GDB, perf, CMake, Kafka, PostgreSQL, Docker, Git, CI/CD